

# HUJI Research Monitor

Healthcare · Weekly Digest · Week 28 · July 06, 2026

## EXECUTIVE SUMMARY

This week's digest showcases Hebrew University's leading-edge innovations across a diverse portfolio, with a strong focus on advanced therapeutic modalities and precision diagnostics. Highlights include groundbreaking CAR T-cell therapy targeting autoimmune diseases, scalable stem cell treatments for neurological disorders, and novel biologics for regenerative medicine. We also feature high-impact medical devices, next-generation companion diagnostics for immunotherapy, and AI-driven solutions poised to accelerate drug discovery and enhance patient care.

#1

## B cell Maturation Antigen Targeted Chimeric Antigen Receptor T Cell Therapy for Refractory Systemic Lupus Erythematosus and Systemic Sclerosis.

37/50

**Main Researcher:** Chamutal Gur

[Immunology] [Clinical] [Synthetic Biology]

*"Next-gen CAR T-cell therapy shifts oncology paradigm to high-value, chronic autoimmune disease management."*

### WHY NOW

First-generation CAR T has proven efficacy in oncology; expanding this advanced therapy to autoimmune diseases opens a vast, underserved market. Early clinical data shows significant promise.

### COMMERCIAL ANGLE

Developing specialized CAR T-cell therapies for autoimmune indications represents a massive shift from oncology toward high-value chronic disease management markets.

<https://europepmc.org/article/MED/42098533>

#2

## Pro-repair properties of a human embryonic stem cell-derived astrocyte cell therapy in demyelinating disorders.

36/50

**Main Researcher:** Tamir Ben-Hur

[Neuroscience] [Immunology] [Clinical] [Synthetic Biology]

*"Off-the-shelf stem cell therapy offers scalable brain repair for multiple sclerosis and demyelinating disorders."*

### WHY NOW

Current MS treatments primarily manage symptoms; a therapy promoting remyelination represents a paradigm shift. The "off-the-shelf" nature addresses scalability, a major hurdle for cell therapies.

### COMMERCIAL ANGLE

Development of a scalable, off-the-shelf cell therapy for remyelination in Multiple Sclerosis and other demyelinating neurological disorders.

<https://pubmed.ncbi.nlm.nih.gov/42276062/>

#3

## The effect of a proprietary chlorine-stabilizing surface-coating formulation on healthcare environment disinfection in multidrug-resistant bacteria isolation rooms.

33/50

**Main Researcher:** Oren Zimhony

[Medical Device] [Materials] [Clinical]

*"Durable antimicrobial surface coating enhances infection control, combating multidrug-resistant bacteria in healthcare settings."*

### WHY NOW

Hospitals face immense pressure to reduce healthcare-associated infections (HAIs) and antibiotic resistance. This scalable, long-lasting solution provides a critical upgrade to existing disinfection protocols.

### COMMERCIAL ANGLE

This technology offers a scalable solution for hospitals and long-term care facilities to enhance infection control protocols and reduce the transmission of multidrug-resistant organisms through high-durability antimicrobial coatings.

<https://pubmed.ncbi.nlm.nih.gov/42374536/>

#4

## Recombinant Human Amelogenin Protein Enhances Healing of Osteochondral Injury in a Goat Model.

33/50

**Main Researcher:** Amir Haze

[Medical Device] [Clinical] [Synthetic Biology] [Proteomics]

*"Recombinant human protein significantly accelerates cartilage and bone healing for osteoarthritis and joint injuries."*

### WHY NOW

With an aging population, the prevalence of osteoarthritis and joint injuries is rising, creating a huge market for regenerative medicine biologics that offer true repair, not just symptom management.

### COMMERCIAL ANGLE

Development of a high-value regenerative medicine biologic for treating osteoarthritis and traumatic joint injuries in humans.

<https://pubmed.ncbi.nlm.nih.gov/41925697/>

#5

## APOBEC mutational signatures predict immune checkpoint inhibitor benefit in TMB-low metastatic NSCLC independent of PD-L1 status.

32/50

**Main Researcher:** N Peled

[Diagnostics] [Genomics] [Immunology] [Software/AI]

*"Novel diagnostic identifies more lung cancer patients for life-saving immunotherapy, expanding market access."*

### WHY NOW

Current immunotherapy selection criteria (PD-L1, TMB) exclude many potential responders. This AI-powered companion diagnostic can broaden patient eligibility, driving market growth for immune checkpoint inhibitors.

### COMMERCIAL ANGLE

Development of a companion diagnostic test using machine learning to identify APOBEC signatures in TMB-low lung cancer patients to expand the market for immune checkpoint inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42378718/>

#6

## Targeting primary and metastatic ovarian cancer with a peptide derived from the human NAF-1/CISD2 protein.

29/50

**Main Researcher:** Rachel Nechushtai

[Drug Discovery] [Clinical] [Immunology]

*"First-in-class peptide therapy targets primary and metastatic ovarian cancer by disrupting tumor growth."*

### WHY NOW

Ovarian cancer remains a deadly disease with significant unmet needs, especially for metastatic forms. A novel therapeutic with a targeted mechanism could represent a major breakthrough and high-value asset.

### COMMERCIAL ANGLE

The peptide represents a high-potential therapeutic candidate for a first-in-class oncology drug, suitable for licensing to pharmaceutical companies focusing on precision cancer therapies or protein-protein interaction inhibitors.

<https://pubmed.ncbi.nlm.nih.gov/42259134/>

#7

## Breath-Based Detection of Oral Diseases Using Sensors and Machine Learning.

28/50

**Main Researcher:** Yael Houri-Haddad

[Medical Device] [Diagnostics] [Software/AI]

*"Non-invasive breathalyzer provides rapid, objective diagnostics for gingivitis and periodontitis in dental clinics."*

### WHY NOW

Demand for convenient, objective diagnostic tools is increasing. This technology offers a significant upgrade over subjective visual assessments, improving early detection and patient management in a large market.

### COMMERCIAL ANGLE

Development of a non-invasive, rapid-screening handheld breathalyzer for dental clinics to provide objective diagnostics for periodontal health.

<https://pubmed.ncbi.nlm.nih.gov/42379259/>

#8

## RadD from *Fusobacterium nucleatum* engages NKp46 to promote antitumor cytotoxicity.

28/50

**Main Researcher:** Ofer Mandelboim

[Immunology] [Drug Discovery] [Clinical]

*"Bacterial protein fragment activates natural killer cells, unlocking a novel immunotherapy pathway for cancer."*

### WHY NOW

NK cell-based immunotherapies are a rapidly growing field, complementing T-cell approaches. Discovering a potent, specific NK cell agonist from a microbial source offers a unique development pathway.

### COMMERCIAL ANGLE

Development of RadD-based NK cell agonists or biomimetic immunotherapies to enhance natural killer cell cytotoxicity in oncology.

<https://europepmc.org/article/MED/42065370>

#9

## Targeting Temozolomide-Resistant Glioblastoma: Therapeutic Potential of Neuronal Nitric Oxide Synthase Inhibitor.

26/50

**Main Researcher:** Haitham Amal

[Drug Discovery] [Neuroscience] [Clinical]

*"Novel inhibitor overcomes drug resistance in deadly glioblastoma, boosting standard chemotherapy effectiveness."*

### WHY NOW

Glioblastoma is notoriously resistant to treatment, leading to poor prognoses. An adjuvant therapy that resensitizes tumors to existing drugs addresses a critical unmet need and offers significant market value.

### COMMERCIAL ANGLE

Development of a high-value adjuvant therapy to extend the clinical utility of existing standard-of-care chemotherapy for resistant brain tumors.

<https://pubmed.ncbi.nlm.nih.gov/42374654/>

#10

## Strategic Template Filtering Accelerates Fragment-Based Peptide Docking.

26/50

**Main Researcher:** Ora Schueler-Furman

[Drug Discovery] [Software/AI] [Proteomics]

*"AI-accelerated peptide docking software slashes drug discovery time and cost, boosting therapeutic design."*

### WHY NOW

Pharmaceutical companies are heavily investing in AI/ML to revolutionize R&D.; This optimized algorithm offers a direct pathway to enhance efficiency in the high-growth area of peptide therapeutics.

### COMMERCIAL ANGLE

Licensing or integrating this high-efficiency docking algorithm into AI-driven drug discovery platforms to accelerate the design of peptide-based therapeutics.

<https://pubmed.ncbi.nlm.nih.gov/42299509/>

#11

## Safety and Efficacy of Direct Selective Laser Trabeculoplasty in Patients With Ocular Hypertension or Open Angle Glaucoma.

26/50

**Main Researcher:** David Zadok

[Medical Device] [Clinical]

*"Laser-based ophthalmic device offers safe, effective first-line treatment for glaucoma, reducing reliance on drops."*

### WHY NOW

Glaucoma affects millions, and patient adherence to daily eye drops is a challenge. A non-invasive, office-based laser procedure as a primary treatment addresses a significant market need for convenience and efficacy.

### COMMERCIAL ANGLE

The findings support the commercialization and expanded clinical adoption of laser-based ophthalmic surgical devices as a first-line alternative to traditional glaucoma eye drops.

<https://europepmc.org/article/MED/42148860>